

$$\begin{aligned}
& \frac{1}{16g^2} \left(-\frac{9}{2} g_2^6 \text{LF}_{3,0} [m_2] - \frac{1}{6} g_2^6 \text{LF}_{4,-1} [m_2] + \frac{8}{45} g_2^6 \text{LF}_{5,-2} [m_2] + \right. \\
& g_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} (g_1^2 - 3 g_2^2) + \sum_p g_2^4 (-9 g_1^2 + 5 g_2^2) \big) \text{LF}_{3,0} [m_1^{\text{p}}] + \\
& \overline{y_e^{\text{pr}}} y_e^{\text{pr}} - \frac{1}{48} \sum_p g_2^6 \big) \text{LF}_{4,-1} [m_1^{\text{p}}] - \frac{2}{45} \sum_p g_2^6 \text{LF}_{5,-2} [m_1^{\text{p}}] + \\
& (-6 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (g_1^2 + 9 g_2^2) - 6 g_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_p g_2^2 (3 g_1^2 + 5 g_2^2)) \text{LF}_{3,0} [m_q^{\text{p}}] + \\
& \overline{y_d^{\text{pr}}} y_d^{\text{pr}} - \frac{1}{16} \sum_p g_2^6 \big) \text{LF}_{4,-1} [m_q^{\text{p}}] - \frac{2}{15} \sum_p g_2^6 \text{LF}_{5,-2} [m_q^{\text{p}}] - \\
& \text{LF}_{3,0} [\tilde{\mu}] - \frac{1}{12} g_2^6 \text{LF}_{4,-1} [\tilde{\mu}] + \frac{4}{45} g_2^6 \text{LF}_{5,-2} [\tilde{\mu}] - g_1^6 \tilde{\mu}^2 \text{LF}_{3,3,-2} [m_1, \tilde{\mu}] - \\
& \tilde{\mu}^2 \text{LF}_{3,3,-1} [m_1, \tilde{\mu}] + \frac{2}{3} g_2^6 \text{LF}_{2,1,0} [m_2, \tilde{\mu}] + \frac{17}{12} g_2^6 \text{LF}_{2,2,-1} [m_2, \tilde{\mu}] - \\
& \text{LF}_{3,1,-1} [m_2, \tilde{\mu}] - \frac{7}{3} g_2^6 \text{LF}_{3,2,-2} [m_2, \tilde{\mu}] + \frac{3}{2} g_2^6 m_2^2 \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] + \\
& \text{LF}_{3,3,-3} [m_2, \tilde{\mu}] - 3 g_2^6 (m_2^2 + \tilde{\mu}^2) \text{LF}_{3,3,-2} [m_2, \tilde{\mu}] - 3 g_2^6 m_2^2 \tilde{\mu}^2 \text{LF}_{3,3,-1} [m_2, \tilde{\mu}] + \\
& \text{LF}_{4,2,-3} [m_2, \tilde{\mu}] - g_2^6 m_2^2 \text{LF}_{4,2,-2} [m_2, \tilde{\mu}] - \frac{1}{2} g_1^2 g_2^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{p}}] + \\
& \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \overline{y_u^{\text{st}}} y_u^{\text{st}} y_u^{\text{pt}} \text{LF}_{2,1,0} [m_d^{\text{r}}, m_u^{\text{t}}] - \frac{1}{2} g_1^2 g_2^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{p}}] - \\
& \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{4} g_1^2 g_2^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,2,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} (3 g_1^2 + 4 g_2^2) - 3 g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] - \\
& (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} (3 g_1^2 + 4 g_2^2) - 3 g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \overline{y_e^{\text{st}}} y_e^{\text{st}} y_e^{\text{sr}} \text{LF}_{2,1,0} [m_l^{\text{p}}, m_l^{\text{s}}] - \frac{1}{2} g_2^2 \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{3,1,-1} [m_l^{\text{p}}, m_l^{\text{s}}] - \\
& \overline{a_d^{\text{pr}}} (2 g_1^2 + 3 g_2^2) \text{LF}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& g_2^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} (5 g_1^2 + 4 g_2^2) - g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] - \\
& (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} (3 g_1^2 + g_2^2) - 3 g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& y_d^{\text{sr}} (12 g_2^2 \overline{y_d^{\text{st}}} y_d^{\text{pt}} + \overline{y_u^{\text{st}}} y_u^{\text{pt}} (g_1^2 + 3 g_2^2)) \text{LF}_{2,1,0} [m_q^{\text{p}}, m_q^{\text{s}}] - \\
& \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{3,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}] + \\
& (3 g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (-g_1^2 + g_2^2)) \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] + \\
& (-g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 3 g_2^2)) \text{LF}_{3,1,0} [m_q^{\text{p}}, m_u^{\text{r}}] - \\
& \frac{1}{2} \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_u^{\text{r}}] - \frac{3}{2} g_2^4 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_u^{\text{r}}] + \\
& y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,0} [m_q^{\text{s}}, m_q^{\text{p}}] + \\
& (7 g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 - g_2^2)) \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] - \\
& (g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + g_2^2)) \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{p}}] + \\
& (-5 g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + g_2^2)) \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] - \\
& (-3 g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] + \\
& y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} (-2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,0} [m_u^{\text{t}}, m_d^{\text{r}}] - \\
& \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1} [m_u^{\text{t}}, m_d^{\text{r}}] + \frac{1}{2} g_1^2 g_2^4 \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] - \\
& g_2^2 \text{LF}_{3,2,-2} [\tilde{\mu}, m_1] - \frac{3}{4} g_1^4 g_2^2 m_1^2 \text{LF}_{3,2,-1} [\tilde{\mu}, m_1] - \frac{7}{12} g_1^2 g_2^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] + \\
& g_2^2 \text{LF}_{4,2,-3} [\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 g_2^2 m_1^2 \text{LF}_{4,2,-2} [\tilde{\mu}, m_1] + \frac{1}{6} g_1^2 g_2^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + \\
& \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] - \frac{65}{12} g_2^6 \text{LF}_{3,2,-2} [\tilde{\mu}, m_2] - \frac{3}{4} g_2^6 m_2^2 \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] - \frac{3}{4} g_2^6 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + \\
& \text{LF}_{4,2,-3} [\tilde{\mu}, m_2] + \frac{1}{2} g_2^6 (m_2^2 + \tilde{\mu}^2) \text{LF}_{4,2,-2} [\tilde{\mu}, m_2] + \frac{1}{2} g_2^6 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] - \\
& g_2^4 \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_1] - \frac{3}{2} g_1^2 g_2^4 (m_1 m_2 + 2 \tilde{\mu}^2) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_1] + \\
& \text{LF}_{3,2,1,-3} [m_2, \tilde{\mu}, m_1] + g_1^2 g_2^4 (m_1 m_2 + 2 \tilde{\mu}^2) \text{LF}_{3,2,1,-2} [m_2, \tilde{\mu}, m_1] - \\
& \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{ur}} \overline{y_u^{\text{uv}}} y_u^{\text{sv}} \text{LF}_{1,1,1,0} [m_d^{\text{r}}, m_d^{\text{t}}, m_u^{\text{v}}] + \\
& (\overline{y_u^{\text{st}}} y_u^{\text{pt}} \over$$